Sardar Md. Saffat Zabin

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Education

Bangladesh University of Engineering & Technology (BUET) — Dhaka, Bangladesh
B.Sc. in Computer Science and Engineering; ***CGPA: 3.59/4.00*** — *Feb 2020 – Mar 2025*
Thesis: Neural Prefetch Filtering for Microarchitectural Prediction
Advisors: Dr. Rifat Shahriyar, Dr. Abdullah Muzahid (Texas A&M)

Notre Dame College — Dhaka, Bangladesh
Higher Secondary Certificate (HSC); ***GPA: 5.00/5.00*** — *2017 – 2019*

Research Interests

Computer Architecture (CPU/GPU Architecture, Hardware prefetching, Memory Hierarchy, Hardware-Software Co-design) · **Database & Storage Systems** (LSM-tree key-value stores) · **Machine Learning for Systems** · **Energy-Efficient Data-Center Computing**

Research Experience

CHiLL (Computing Hardware & Large Data Lab), University of Massachusetts Boston — Remote
Graduate Research Intern; Advisor: Dr. Tarikul Islam Papon — *Jan 2026 – Present*

- Designing a **learned compaction trigger** for RocksDB-based LSM-tree key-value stores: a reinforcement learning controller decides, level by level, whether to compact early or defer by adjusting the L0 file-count trigger and per-level size targets, while RocksDB keeps full control of file selection and correctness.
- Uses an analytic LSM cost model as a prior with a learned residual on top, so the policy starts cold and adapts online to workload shifts, without offline training or tuning sweeps; the goal is lower read and write amplification than stock RocksDB.
- Implementing the controller inside RocksDB in **C++**, with training in Python; evaluating with db_bench on bare-metal Chameleon Cloud nodes, using the Facebook production workload model of Cao et al. (FAST 2020).

Independent Research Collaboration — Dhaka, Bangladesh
Researcher, Vision–Language Models; Advisor: Dr. M. Saifur Rahman — *Oct 2025 – Jan 2026*

- Built a **compute-efficient Bengali image-captioning pipeline**: a frozen MaxViT encoder, a Bengali-native mBART-50 decoder and a lightweight bridge between them, trained on LaBSE-verified English–Bengali pairs and 110k bilingual-prompted synthetic images.
- Proposed a three-part training objective (cross-attention-guided Patch-Alignment Loss, InfoNCE and Sinkhorn optimal transport) that improves visual grounding and outperforms strong cross-entropy baselines on Flickr30k-1k (**BLEU-4 12.29, METEOR 27.98, BERTScore-F1 71.20**) and MSCOCO-1k, while narrowing the real–synthetic centroid gap by 41%.
- Outcome: co-first-author paper accepted at **PRICAI 2026**.

Bangladesh University of Engineering & Technology (BUET) — Dhaka, Bangladesh
Undergraduate Thesis Advisee, Computer Architecture; Advisors: Dr. Rifat Shahriyar, Dr. Abdullah Muzahid (Texas A&M) — *Nov 2023 – Mar 2025*

- Reimplemented the perceptron-based prefetch filter of Bhatia et al. (*Perceptron-based Prefetch Filtering*, ISCA 2019) from scratch in **C++** on top of the Signature Path Prefetcher, and compared its coverage, accuracy and IPC with the published results in the ChampSim simulator on SPEC CPU2006 and CPU2017 traces.
- Proposed a **neural-network prefetch filter** (softmax output, cross-entropy loss, online weight updates) that removes the perceptron's one-hot weight tables, cutting weight storage by **88% (13.8 KB to 1.7 KB)** and total filter storage by **35%**, with **0.9%** geometric-mean IPC loss against the perceptron filter.
- Funded by the BUET RISE Undergraduate Student Researcher Grant.

Bangladesh University of Engineering & Technology (BUET) — Dhaka, Bangladesh
Undergraduate Research Assistant, HCI; Advisors: Dr. A. B. M. Alim Al Islam, Dr. Novia Nurain — *Sep 2024 – Mar 2025*

- Studied how the government-imposed Internet shutdown during Bangladesh's July 2024 protests affected citizens, and how they adapted, through long-form interviews analyzed qualitatively and with sentiment analysis.
- Built the analysis pipeline: **Whisper** for Bangla-to-English speech translation, **pyannote** for speaker diarization of 2–3-speaker interviews, and a RoBERTa classifier trained on GoEmotions (27 emotion categories) alongside a DistilRoBERTa emotion classifier to measure emotions in the transcripts.
- Outcome: co-first-author paper at **CHI EA '25** and a poster at NSysS 2024.

Publications

* Equal contribution.

- Riad Ahmed Anonto*, **Sardar Md. Saffat Zabin***, M. Saifur Rahman. Align Where the Words Look: Cross-Attention-Guided Patch Alignment with Contrastive and Transport Regularization for Bengali Captioning. Accepted at the *Pacific Rim International Conference on Artificial Intelligence (**PRICAI 2026**)*; not included in the proceedings, as registration could not be funded. [*arXiv*]
- Farhan Tahmidul Karim*, **Sardar Md. Saffat Zabin***, Ishika Tarin, Novia Nurain, A. B. M. Alim Al Islam. Beyond the Blackout: Qualitative Insights and Sentiment Analysis on the Impact, Adaptation, and Design Solutions During the Internet Shutdown in Bangladesh. In *Extended Abstracts of the CHI Conference on Human Factors in Computing Systems (**CHI EA '25**)*. Acceptance rate 32.83%. [*Paper*]

Posters

- Poster on the Internet-shutdown study (Human–Computer Interaction), *NSysS 2024*.

Industry Experience

Sazim Tech Ltd. Dhaka, Bangladesh
Software Engineer (L-2) *June 2026 – Oct 2026*

- Built an **event-driven**, multi-stage LLM pipeline that scores job candidates' CVs against a job description, with background workers, failure detection, a reconciler for automatic retries and live progress over server-sent events (NestJS, BullMQ, PostgreSQL, LangChain.js).
- Developed backend APIs (Spring Boot, PostgreSQL, Redis) for a Saudi Arabia-based platform, integrating Salesforce for support ticketing and Lean Technologies for IBAN-ownership and business verification; rolled out features behind PostHog feature flags.

OPlusDojo (Remote) Toronto, Canada
Software Engineer *Sep 2025 – May 2026*

- Reduced **p95 API latency from 1851 ms to 30 ms** by restructuring endpoints over PostgreSQL/pgvector and S3 presigned URLs.
- Built an asynchronous document-ingestion and retrieval-augmented generation (RAG) pipeline (FastAPI, Celery, Redis, pgvector, Gemini); set up CI/CD with GitHub Actions and AWS deployment behind load balancers and CloudFront.

Apurba Technologies Ltd. Dhaka, Bangladesh
Full Stack Developer (L-1) *Mar 2025 – Oct 2025*

- Built a concurrent converter from legacy ASCII Bangla fonts to Unicode (JavaFX) and extended it into a Spring Boot and Angular web platform, deployed on a government server (ocr.bangla.gov.bd).

Honors & Awards

- **RISE Undergraduate Student Researcher Grant**, Research & Innovation Center for Science & Engineering (RISE), BUET. Competitive grant funding the implementation of promising undergraduate thesis proposals (Nov 2023 – Mar 2025).

Selected Projects

- **xv6 Operating System Extensions** (C) [*GitHub*]: added system calls, inter-process communication, copy-on-write and paging to the xv6 teaching kernel.
- **Processor Design in Logisim** [*GitHub*]: built a 32-bit floating-point adder/subtractor and a 4-bit MIPS-style pipelined processor from basic gates and ICs.
- **C Compiler** (C) [*GitHub*]: incremental compiler for a subset of the C language.
- **Embedded Game on ATmega32A** [*GitHub*]: game driven by an ATmega32A and an Arduino UNO, with a 16×16 LED matrix built from four 8×8 matrices and decoder ICs, four joystick-controlled paddles and an MPU-6050 gyroscope.
- **Computer Networks** (C++, Java) [*GitHub*]: implemented TCP-Reno congestion control and network topologies in ns-3; socket programming in Java.
- **Machine Learning from Scratch** (Python) [*GitHub*]: implemented logistic regression, bagging and stacking ensembles, a feed-forward neural network, EM and PCA from scratch.
- **UrbanNest: Real-Estate Platform** (Spring Boot, React, AWS): full-stack web platform; cut database queries per page from 81+ to 2–3 by removing N+1 query patterns.

Relevant Coursework

Computer Science: Computer Architecture, Microprocessors, Digital Logic Design, Operating Systems, Database Systems, Compilers, Computer Networks, Data Structures & Algorithms, Algorithm Engineering, Theory of Computation, Machine Learning, Artificial Intelligence

Mathematics: Linear Algebra, Probability, Statistics, Vector Calculus, ODE, PDE, Complex Variables, Fourier Analysis, Laplace Transformation

Technical Skills

Programming Languages: C/C++, Python, Java, TypeScript, SQL

Research Tools & Systems: ChampSim, RocksDB (db_bench), Chameleon Cloud, PyTorch, Hugging Face Transformers, ns-3, xv6, Linux, LaTeX

Software Engineering: Spring Boot, FastAPI, NestJS, React, PostgreSQL (pgvector), Redis, Celery, BullMQ, Docker, GitHub Actions, AWS (EC2, RDS, S3, CloudFront)

Service & Leadership

Reviewer, PRICAI 2026 2026

Reviewed 3 submitted papers for the Pacific Rim International Conference on Artificial Intelligence

IEEE Computer Society BUET Student Branch Chapter Dhaka, Bangladesh

Publication Committee Chair *Mar 2024 – Mar 2025*

Assistant Treasurer *Mar 2023 – Mar 2024*

Programming Contest Coordinator *Feb 2022 – Mar 2023*

References

Dr. Tarikul Islam Papon
Assistant Professor, Department of Computer Science
University of Massachusetts Boston
Director, CHiLL Lab
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Relation: Research Advisor

Dr. Rifat Shahriyar
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rifat.shahriyar@gmail.com
Relation: Undergraduate Thesis Advisor, Research Advisor